Day 5: Tasks

1. **Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.**
   ➢

```
C:\Users\Disha>ping 192.168.31.108

Pinging 192.168.31.108 with 32 bytes of data:
Reply from 192.168.31.108: bytes=32 time=583ms TTL=64
Reply from 192.168.31.108: bytes=32 time=541ms TTL=64
Reply from 192.168.31.108: bytes=32 time=280ms TTL=64
Reply from 192.168.31.108: bytes=32 time=315ms TTL=64

Ping statistics for 192.168.31.108:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 280ms, Maximum = 583ms, Average = 429ms
```

   - The laptop and phone were connected to the same Wi-Fi network. A ping test was performed from the laptop to the phone's IP address, and reply messages were received successfully.

2. **Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.**
   ➢

```
Tracing route to youtube-ui.l.google.com [2404:6800:4009:820::200e]
over a maximum of 30 hops:

  1     5 ms     4 ms     4 ms  jiofiber.local.html [2409:4090:106b:f811:9632:51ff:fe58:4
  2     *        *        *     Request timed out.
  3   340 ms    50 ms   103 ms  2405:203:400:100:172:31:3:164
  4   326 ms   304 ms    19 ms  2405:200:801:b00::f3a
  5   345 ms   356 ms   568 ms  2405:200:809:1510:61::7
  6     *        *        *     Request timed out.
  7   171 ms    48 ms   268 ms  2405:200:801:b00::dfc
  8    80 ms   204 ms   277 ms  2405:200:801:b00::dfd
  9   149 ms   173 ms   267 ms  2404:6800:8285:200::1
 10   628 ms   305 ms   348 ms  2404:6800:8285:200::1
 11   111 ms   340 ms    53 ms  2404:6800:8285:200::1
 12   364 ms    82 ms   318 ms  2001:4860:0:1::2b6e
 13   228 ms   271 ms   471 ms  2001:4860:0:1::1057
 14   393 ms   102 ms    33 ms  bom07s30-in-x0e.1e100.net [2404:6800:4009:820::200e]
```

3. **Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.**
   - **error message I have receive .**

     ```
     C:\Users\Disha>ping [Phone_IP_Address]
     Ping request could not find host [Phone_IP_Address]. Please check the name and try again.
     ```

     After disconnecting the phone from the Wi-Fi network, the laptop no longer received any reply from the phone. The ping command displayed "Request timed out" messages.

4. **Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.**
   -

     ```
     Interface: 192.168.56.1 --- 0x8
       Internet Address      Physical Address      Type
       192.168.56.255        ff-ff-ff-ff-ff-ff     static
       224.0.0.22            01-00-5e-00-00-16     static
       224.0.0.251           01-00-5e-00-00-fb     static
       224.0.0.252           01-00-5e-00-00-fc     static
       232.24.3.241          01-00-5e-18-03-f1     static
       239.255.255.250       01-00-5e-7f-ff-fa     static

     Interface: 192.168.31.87 --- 0xb
       Internet Address      Physical Address      Type
       192.168.31.1          94-32-51-58-48-0b     dynamic
       192.168.31.108        4e-e6-7c-57-e6-ea     dynamic
       192.168.31.255        ff-ff-ff-ff-ff-ff     static
       224.0.0.22            01-00-5e-00-00-16     static
       224.0.0.251           01-00-5e-00-00-fb     static
       224.0.0.252           01-00-5e-00-00-fc     static
       232.24.3.241          01-00-5e-18-03-f1     static
       239.255.255.102.18    01-00-5e-7f-66-12     static
       239.255.255.250       01-00-5e-7f-ff-fa     static
       255.255.255.255       ff-ff-ff-ff-ff-ff     static
     ```

5. **Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.**

Network Setup and Connectivity Testing Report

6. I connected my Windows laptop and an Android phone to the same Wi-Fi network. After finding the phone's IP address, I used the Ping command in Command Prompt to test communication between the devices.
7. Command used:
   ping [Phone IP Address]
8. The ping test was successful, and reply messages were received, confirming that both devices could communicate on the network.
9. Next, I used the Traceroute command to check the path to YouTube.
10. Command used:
    tracert www.youtube.com
11. The command showed several hops between my device and YouTube's server, confirming internet connectivity.
12. I then disconnected the phone from Wi-Fi and ran the ping command again. This time, the message "Request timed out" appeared because the phone was no longer connected to the network.
13. Finally, I used the `arp -a` command to view devices connected to the local network and identify their IP addresses.
14. **Conclusion:**
    The Ping, Traceroute, and ARP commands helped me test network connectivity, view routing information, and identify devices connected to the network.